20 In London, 1 litre of petrol costs 108.9p
In New York, 1 US gallon of petrol costs $2.83

1 US gallon = 3.785 litres
£1 = $1.46

In which city is petrol better value for money, London or New York?
You must show your working.

Compare the cost in pounds of one US gallon.

London: 108.9p = £1.089 per litre.

L	GBP
1	1.089
3.785	4.121865

Multiply both columns by 3.785.

1 mark
Convert the volume unit

1.089 × 3.785 = £4.121865

New York: convert $2.83 into pounds.

USD	GBP
1.46	1
2.83	$\frac{283}{146}$

$\times\frac{283}{146}$ $\times\frac{283}{146}$

1 mark
Use comparable currency and volume

2.83 ÷ 1.46 = £1.938356...

New York is cheaper: £1.94 < £4.12.

1 mark
Correct city with comparable costs

(Total for Question 20 is 3 marks)

21 A gold bar has a mass of 12.5 kg.

The density of gold is 19.3 g/cm^3

Work out the volume of the gold bar.
Give your answer correct to 3 significant figures.

..... cm^3

(Total for Question 21 is 3 marks)

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